LIMEN Helix — Verified Study Reference Set (Phases 0–10)

Compiled: July 8, 2026 **Source of verification:** All records confirmed via PubMed (`lookup_article_by_citation` → `get_article_metadata`, with metadata disambiguation on every ambiguous return). **Scope:** The 11-phase spiral / hub-spoke connectome. This is the verified spine only. Nothing here is included unless its existence, authorship, journal, and DOI were confirmed this session.

How to read this document

Two axes are kept separate on purpose, because collapsing them is the core failure mode this project guards against:

- **Verified-real** = the paper exists and says what it's attributed to say (an existence-and-attribution check).
- **Established-true** = the claim the paper supports is itself settled science (a truth check).

Every entry below is verified-real. The **confidence tier** speaks only to *established-true* — how much weight the claim can bear — and it is NOT raised by the fact that the DOI checks out. A verified DOI never launders a hypothesis into a finding.

Confidence tiers

Tier	**Meaning**
SOLID	Consensus mechanism; well-supported.
STRONG	Strong primary/meta-analytic evidence.
MIXED	Cited, but with a labeled seam — a contested sub-claim or an interpretive layer riding on top.
HYPOTHESIS	Real, influential, but framed by its own authors as a testable model, not settled fact.
INTERPRETIVE / UNSUPPORTED	No verified mechanistic primary. Honest label required; renders as framing, not evidence.

Cross-cutting anchor — Precision-weighting spine

These four run underneath the whole spiral. The interpretive claim that the phases sit on a single precision-weighting axis (a representation checked upstream by an invalidator) *borrows* from this spine; it is a synthesis, not a claim any one of these papers makes on its own.

#	Citation	Tier	Note
1	Rao & Ballard (1999)	SOLID	Predictive coding: feedback carries prediction, feedforward carries error. The map/territory relation in cortical wiring.
2	Friston (2010)	MIXED	Free-energy principle. Influential unifying framework; sweeping by nature. (Verified as the *review*, not the reply-letter it was ambiguous with.)
3	Carhart-Harris & Friston (2019)	HYPOTHESIS	REBUS. States the anchor in its own words: relaxation of high-level prior precision. Load-bearing for Phase 7.
4	Menon (2011)	STRONG (for triple-network machinery)	Triple-network model. **Supports the switching machinery only. Does NOT describe reintegration/"resur**

#	Citation	Tier	Note
			rection" — see Phase 10.

Phase-by-phase

Phase 0 — SOURCE (DMN baseline self-referential processing)

Tier: SOLID (as DMN identification) — but see gap. No dedicated Phase-0 primary was verified this session. The DMN core-node identification it relies on is carried by papers verified for other phases — Brewer 2011 (#24, mPFC + PCC as DMN core) and Sheline 2009 (#12, DMN self-referential activity). **Gap flag:** The canonical DMN baseline primary (e.g. Raichle 2001 / Buckner 2008) was *not* verified here. Phase 0 should either get its own verified anchor or explicitly borrow #24/#12.

Phase 1 — RUPTURE (salience switching / prediction error)

Tier: STRONG. Primary causal support: **Sridharan 2008 (#5)** — right fronto-insular cortex causally drives the CEN↔DMN switch (Granger-causality + chronometric, replicated across three paradigms). Dissociation: **Seeley 2007 (#6)**. Precision-weighting reading of the switch rides on #1/#2 — interpretive layer, not asserted by #5/#6.

> **Disambiguation note:** the wrong Sridharan candidate was an induced-pluripotent-stem-cell paper by a different R. Sridharan. Caught at metadata.

Phase 2 — RHYTHM (vagal tone / mirror systems / co-regulation)

Tier: MIXED. Autonomic mechanism: **Thayer 2000 (#13), Thayer 2012 (#14)** — SOLID. Mirror system: **Rizzolatti & Craighero 2004 (#17)** — existence SOLID, but the leap to empathy/co-regulation *function* is contested (Hickok critique). Co-regulation/synchrony: **Feldman 2017 (#18)** — solid but largely parent–infant/attachment population; adult transformational co-regulation is an extension. "Ventral vagal social engagement" language: **Porges 2007 (#15) — CONTESTED.**

Phase 3 — DARKNESS (threat circuitry / DMN dysregulation)

Tier: MIXED (mechanism strong as clinical findings; identity with a transformational phase is a labeled prediction).

- *Extinction edge (vmPFC→amygdala, inhibitory):* **Milad & Quirk 2002 (#7), Phelps 2004 (#8), Milad & Quirk 2012 (#9).** Direction and sign primary-sourced. Rodent IL→human vmPFC is a homology, not identity.
- *DMN dysregulation:* **Kaiser 2015 (#10), Hamilton 2015 (#11), Sheline 2009 (#12).** Precise reliable finding is *elevated DMN–subgenual-PFC coupling + failure to down-regulate DMN* — not a blanket "DMN hyperconnected."
- *Phenomenology:* **Lindahl & Britton 2017 (#16)** — documents challenging/adverse contemplative states as a real catalogued category. Descriptive; no imaging.
- **Seam:** all mechanism papers are *clinical depression*. Mapping onto a transient transformational dark phase is a hypothesis (medium-low identity), assigned to the Tier-3 EEG lane to test.

Phase 4 — PEACE (vmPFC top-down autonomic control)

Tier: STRONG (mechanism); polyvagal sub-claim CONTESTED. Thayer 2000 (#13), Thayer 2012 (#14) — HRV indexes vmPFC/amygdala top-down control. Note: neurovisceral integration (Thayer) and polyvagal (Porges #15) are *different* frameworks; Thayer carries the mechanistic weight, Porges supplies the contested "ventral vagal" label. Same inhibitory vmPFC→amygdala edge as Phase 3, opposite state.

Phase 5 — AWARENESS (salience switching + metacognition)

Tier: STRONG (both components), but two distinct substrates. Switching: **Sridharan (#5), Seeley (#6)** (aPFC-independent). Metacognition: **Fleming 2010 (#19), Fleming & Dolan 2012 (#20)** — introspective accuracy tracks anterior-PFC structure, dissociable from task performance. These are *different regions*; the phase label bundles them and the row should show both. Metacognition = second-order confidence estimate = precision applied to the self-model (interpretive bridge to Phase 8).

Phase 6 — ORDER (executive control / DLPFC)

Tier: SOLID. Miller & Cohen 2001 (#23) — PFC maintains goal representations that bias processing. Titled a "theory," but consensus scaffolding, not a fringe proposal (contrast Phase 7).

Phase 7 — DISSOLUTION (entropic brain / relaxed priors)

Tier: HYPOTHESIS (by the authors' own framing). Primary: **Carhart-Harris 2014 (#21), 2018 revisit (#22).** Mechanism sibling: **REBUS 2019 (#3).**

- **Recommended split:** mechanism = REBUS/precision-relaxation (#3), the narrower, more falsifiable claim; associated phenomenology = entropic-brain criticality (#21/#22), flagged-contested and explicitly *not* load-bearing. Prevents the weaker criticality claim laundering into the row's tier.

- **Seam:** both #21 and #3 are *psychedelic-state* studies. Mapping onto a drug-free transformational dissolution is a labeled prediction, same class as the Phase-3 clinical seam.

Phase 8 — WITNESS (decentering / PCC reconfiguration)

Tier: MIXED. Brewer 2011 (#24) — experienced meditators show relative deactivation of DMN core (mPFC + PCC) and stronger PCC↔DLPFC/dACC coupling. Mechanism solid; "decentering" as the subjective construct is inferred from practice condition, not directly measured. **Double duty:** #24 is also the empirical anchor that the DMN hub is a real, state-dependent, measurable object, and the PCC↔DLPFC coupling it reports is a WITNESS↔ORDER (P8↔P6) inter-spoke edge with primary support.

Phase 9 — THRESHOLD (gamma coherence / cross-network integration)

Tier: MIXED (robust correlate; "gamma = integration" contested). Melloni 2007 (#25) — perceived stimuli induce transient long-distance gamma-phase synchrony; unperceived do not. Shows a correlate; does **not** settle binding-by-synchrony (attention/microsaccade confounds; debate unresolved). Contested status load-bearing on this row, as with Phase 7. (THRESHOLD = LIMEN; keep the name.)

Phase 10 — RESURRECTION (narrative reconstruction / identity consolidation)

Tier: INTERPRETIVE / UNSUPPORTED-BY-VERIFIED-PRIMARY. No verified mechanistic primary exists for this phase. Menon 2011 (#4) is a triple-network *psychopathology* review and does **not** describe reintegration, recursive rebirth, or cross-domain plasticity; citing it for a phase-10 flourishing state is misattribution. Narrative-identity reconstruction is a real research area, but mapping it onto a discrete terminal "rebirth" phase is interpretive. **Required handling:** render as framing, not evidence. The prediction that reintegration involves sustained cross-network coherence (DMN–CEN) is a legitimate, *untested* hypothesis riding on the switching literature — label it as such. **Do not attach "fMRI confirmed" language to any single-client vignette.**

Full reference list (verified this session)

All DOIs confirmed via PubMed. Format: Author (Year). Title. Journal Vol(Issue):Pages. PMID. DOI.

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12. Sheline YI, Barch DM, Price JL, Rundle MM, Vaishnavi SN, Snyder AZ, Mintun MA, Wang S, Coalson RS, Raichle ME (2009). The default mode network and self-referential processes in depression. *Proc Natl Acad Sci U S A* 106(6):1942–1947. PMID 19171889. https://doi.org/10.1073/pnas.0812686106
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Coverage summary

Phase	Name	Verified anchor(s)	Tier
0	SOURCE	#24, #12 (borrowed; no dedicated primary)	SOLID* / gap

Phase	Name	Verified anchor(s)	Tier
1	RUPTURE	#5, #6 (+#1, #2)	STRONG
2	RHYTHM	#13, #14, #17, #18, #15	MIXED
3	DARKNESS	#7, #8, #9, #10, #11, #12, #16	MIXED
4	PEACE	#13, #14, #15	STRONG / contested sub-claim
5	AWARENESS	#5, #6, #19, #20	STRONG (2 substrates)
6	ORDER	#23	SOLID
7	DISSOLUTION	#3, #21, #22	HYPOTHESIS
8	WITNESS	#24	MIXED
9	THRESHOLD	#25	MIXED
10	RESURRECTION	*(none verified)*	INTERPRETIVE / UNSUPPORTED

Straight-through status: every phase 0–10 now carries at least one verified citation OR an explicit honest label. No phase is silent; no phase is overclaimed. Two phases (7, 10) will not rise above hypothesis/interpretive on current science — that is a property of the literature, not a coverage gap. Two housekeeping gaps remain: Phase 0 wants its own DMN primary; Phase 10 wants an honest interpretive label rather than a borrowed citation.

Appendix — Excluded / failed verification (audit trail)

Recorded so these cannot re-enter as support later. *Nothing frozen or deleted; the auditor is inside the corpus.*

Excluded as miscategorized:

- Jiang Q, … Beier KT, … Kwan AC (2025). Psilocybin triggers an activity-dependent rewiring of large-scale cortical networks. *Cell*. https://doi.org/10.1016/j.cell.2025.11.009

— real, but a **mouse rabies-tracing plasticity** paper. Belongs in the neuroplasticity spoke with a rodent flag, NOT in Phase 7 / psychedelic-state. (Resolved the "Beltran"→**Beier** confusion.) Not cited per builder decision. *Read from PubMed record only; post-dates training cutoff — full-text check required before any use.*

Children's-site (recursivelove.com) Phase-10 citations — checked, failed as attributed:

- Gillespie-Smith et al. (2024) — real paper is about **camouflaging predicting eating-disorder symptoms** (*Autism* 28(11):2858–2868, https://doi.org/10.1177/13623613241245749). Cited for "recursive recovery patterns." **Misattributed** (also: she is last author, not first).
- "Uljarević, Hedley, & Foley (2019)" — **no matching record.** Real 2019 Uljarević/Hedley papers concern 22q11.2 deletion syndrome and workplace well-being; **"Foley" is not an author on any.** Citation does not exist as written.
- Chaidez et al. (2014) — real paper is **GI problems in autistic children** (*J Autism Dev Disord* 44(5):1117–1127, https://doi.org/10.1007/s10803-013-1973-x). Cited for "warm soup is grounding." **Misattributed** (paper is about GI pathology).
- Parletta et al. (2017) — real, but about **adult depression** (Mediterranean diet + fish oil RCT, https://doi.org/10.1080/1028415X.2017.1411320). Cited for neurodivergent children's recovery foods. **Wrong population.**
- Mazahery et al. (2017) — real and on-topic (omega-3/ASD meta-analysis, https://doi.org/10.3390/nu9020155), but concludes omega-3 *may potentially* help *some* symptoms from 4 small RCTs (n=107) and calls for larger studies. **Overstated in use.**
- Ben-Sasson et al. (2009) — real **sensory-modulation** meta-analysis (https://doi.org/10.1007/s10803-008-0593-3). Cited for "recovery foods." **Unrelated claim.**
- Fossum et al. (2024) — **not verified this session; treat as unconfirmed until checked.**

Pattern: across the children's-site Phase-10 bibliography checked, the failure rate was effectively total — real papers pulled by author/keyword and welded to claims they do not make, plus one non-existent citation. That bibliography cannot be relied on as evidence and must be rebuilt from verified sources, not patched.

Verified-real is not established-true. A confirmed DOI is an existence check, never a truth check. Tiers reflect what each claim can bear. — LIMEN Helix citation protocol.